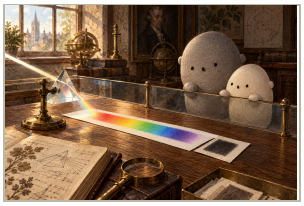
A little strip of light

The light we cannot see. · Curiosity day

Name _____ Date _____



A simplified reconstruction of Ritter's discovery, not a home experiment. The dark sample is a clue to unseen UV.

A SIMPLIFIED STRIP · POSITIONS ONLY

Light	UV-A	Violet	Blue	Green	Yellow	Orange	Red
Place	1	2	3	4	5	6	7

UV-A is unseen. The other six names are visible colors. Real colors blend; these boxes are not equal measured bands.

1. There are 7 places. One is unseen UV-A. How many places name visible colors?

7 − 1 = _____ Show your thinking.

2. Start at place 1, UV-A. How many one-place hops reach place 7, red?

7 − 1 = _____ Show your thinking.

3. Green is at place 4. Red is at place 7. How many places farther along is red?

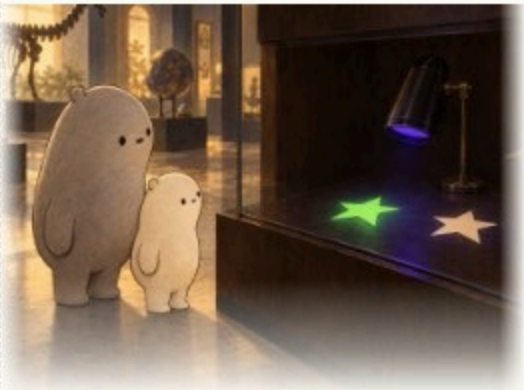
7 − 4 = _____ Show your thinking.

Wavelength is the space from one wave crest to the next. Imagine how stretched out a drawn wave looks. Along our strip toward red, wavelengths get longer. Which end would have more closely spaced crests?

WONDER DAILY

The light we cannot see.

The light we cannot see



One leaf. Six steps. A story that keeps going.

GRADE 2

A green star shines in a quiet museum case. Beside it, a cream star stays dull. Both sit under the same dark-looking lamp. "Is one full of tiny lights?" Little asks. Big leans closer. The two stars have left them a puzzle.



Their search begins long ago. In 1801, Johann Ritter spread sunlight into a rainbow. Light-sensitive material darkened just beyond violet, where he could see no light. An unseen kind of light was there! Today we call it ultraviolet, or UV. "A clue past the rainbow," Little says.



Another display tells of George Stokes. In 1852, he studied materials that took in light and gave out a different color. He called the glow fluorescence. A crystal gleams blue in the case. "Could our star be doing that?" Little wonders. The old discoveries offer a new clue.



Back at the stars, Little studies the black light. It gives out mostly UV-A, the ultraviolet near visible violet. Our eyes cannot see UV-A. A little visible violet makes the lamp look purple. "It isn't sending out darkness," Little says. "It's sending light we cannot see!"



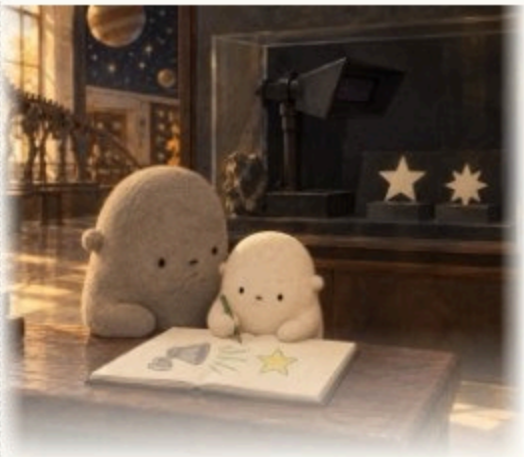
Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



Pictures are simplified and not to scale. Look closely, wonder, and pause whenever you like.

The museum lamp switches off. The green fluorescence stops almost at once. Little sketches a lamp, a star, and light coming out. "No tiny lights inside," Little says. "A material makes the re-glow." Big leaves room on the page. What other material might hold a clue?



The green star has a special coating: a fluorescent phosphor. It absorbs UV-A and quickly emits green light. Absorb means take in; emit means give out. That re-glow is fluorescence. The plain star lacks this coating. "Same shape, different material!" Little says. The glow comes from the star itself.



Give the glow a path.

The light we cannot see. · Curiosity day

Name _____ Date _____

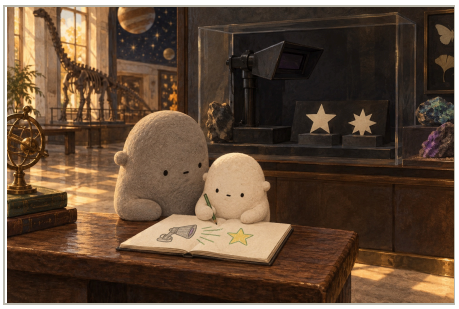
Choose one scene. Say what the clue helped Little discover. Draw, label, or write one idea.



Beyond violet



Light in, light out



Keep the clue

Draw a clue from your scene. Add an arrow to show where light goes or a label to name a change.

UV-A · phosphor · absorb · emit · fluorescence · visible · clue

The dark patch was a clue because … What was reaching it?

The coating takes in … What light goes in? What light comes out?

When the lamp goes off, … What happens to fluorescence?

Point to the detail your words explain. If you write a sentence, check its first capital and ending mark.